

Digital Communications

Baseband Pulse Transmission



Baseband / Passband

- Baseband transmission consists of transmitting the signal without any modulation
- Bandpass (or channel band) transmission consists of shifting frequencies from the baseband to a band of higher frequencies using modulation
- Baseband transmission can be analog or digital
 - Transmission through the atmosphere is limited, as low frequencies will be mixed with other baseband frequencies, and the size of the antennas is unbearable
 - In digital format, line coding is used, resulting in a pulsed signal or a signal with Pulse Amplitude Modulation (PAM)
 - e.g. fiber optic cables and short-range copper links



Baseband / Passband

- In bandpass transmission, digital modulation methods are used so that only a limited frequency band is used
 - is typically used in wireless communications and on channels with bandpass filtering such as POTS
 - allows for Frequency Division Multiplexing (FDM)
- The digital bit stream is first converted to an equivalent baseband signal and then to an RF signal
- On the receiver side, a demodulator is used to detect the signal and reverse the modulation process
 - A combined equipment for modulation and demodulation is called a *modem*

Digital Baseband Transmission

- Baseband pulse transmission requires that the channel be of the low-pass type but with a bandwidth sufficient to accommodate the frequency content of the signals to be transmitted
- However, the frequency response of a channel is of the dispersive type – different from the response of an LPF
- Thus, the result of transmission is that each transmitted pulse is affected by the adjacent pulses
 - ISI – *Intersymbolic Interference*
 - Biggest source of errors
- To correct the ISI it is necessary to correct the shape of the pulses along the entire transmission system – *Pulse Shaping*



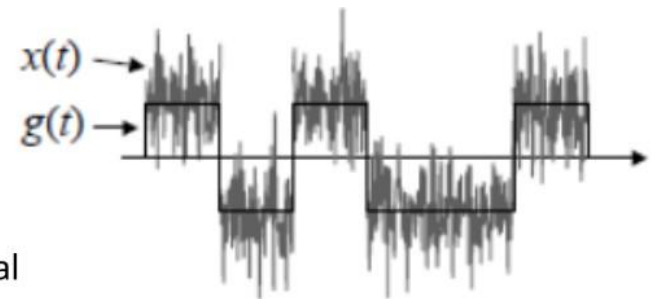
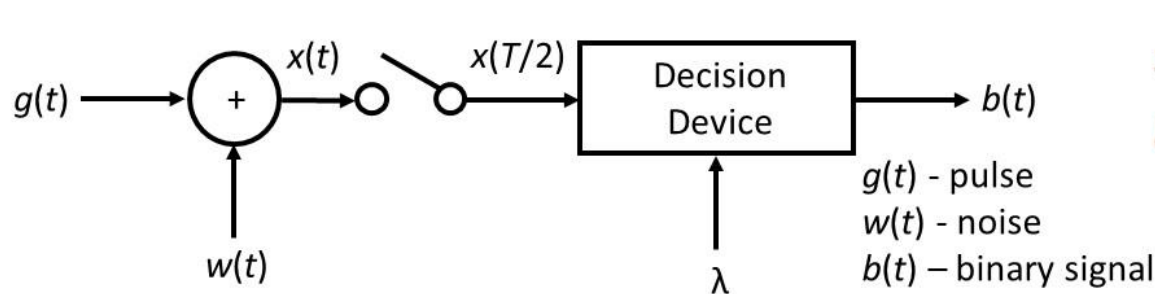
Digital Baseband Transmission

- Another source of errors is the *channel noise*
- Noise and ISI occur simultaneously across the entire channel
- However, to understand how each one affects the performance of the system, they will be considered separately
- One of the main results of telecommunications theory is the detection of pulses with a known waveform contaminated by additive white noise
- The device that allows optimal pulse detection is the so-called *matched filter*
 - Its name derives from the fact that its impulse response is related to the shape of the pulse



Noise – Detection at the center point

- It is possible to remove noise by sampling the signal, for example, at the central point of the pulse, and deciding on the received level
 - This process is called regeneration
- However, there is a certain probability of error that depends on the power of the noise



Noise – Detection at the center point

- In digital communications, the quality of system performance is measured by the probability of error in reception
- To assess the effect of noise on the error probability, consider that the signal at reception has the following form:
 - $x(t) = g(t) + w(t) \quad 0 \leq t \leq T$
 - where $g(t)$ corresponds to a pulse, $w(t)$ is white noise with zero mean and variance σ^2 and T is the transmission period of a symbol
- The receiver samples the signal at time $T/2$ and, based on the sampled value, decides which symbol it is receiving
- If we are transmitting a binary signal with rectangular pulses of amplitude A and $-A$, for the transmission of symbols “1” and “0” respectively, the receiver compares the sample value with a decision level, for example $\lambda = 0$



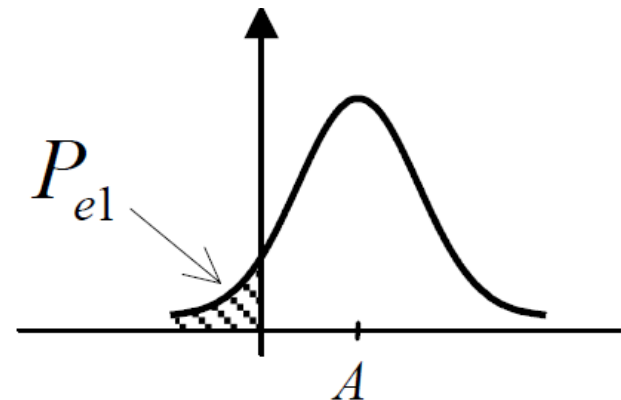
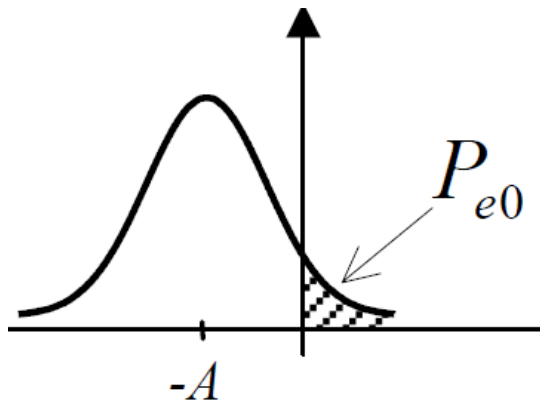
Noise – Detection at the center point

- When the sample has an amplitude greater than λ the receiver decides it has received a “1” and when the sample has an amplitude less than λ the receiver decides it has received a “0”
- When a “1” is transmitted, the random variable $x(T/2)$ has mean A and when a “0” is transmitted, the random variable $x(T/2)$ has mean $-A$
- If the noise is Gaussian and for $\lambda = 0$, the error probability when transmitting a “0” is called P_{e0} , and the error probability when transmitting a “1” is called P_{e1}



Noise – Detection at the center point

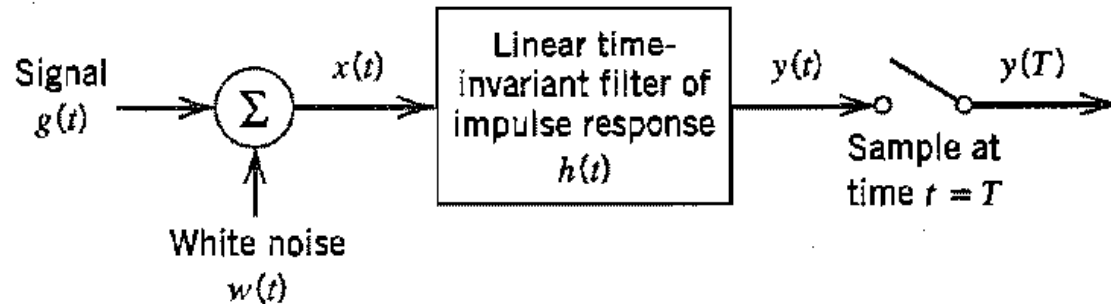
- The probability of error is equal to:
 - $P_e = P_0 \cdot P_{e0} + P_1 \cdot P_{e1}$
 - where P_0 is the probability of a “0” occurring and P_1 is the probability of a “1” occurring



- If $P_0 = P_1$ the optimal decision level corresponds to the average value of the level of the two pulses, which in the example we are considering is $[A + (-A)]/2 = 0$, that is, in this case $\lambda = 0$ is the optimal decision level

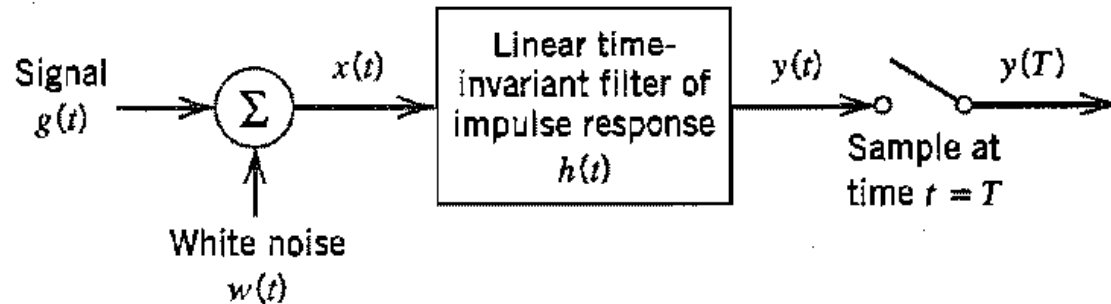
Matched Filter

- Let us consider the model of a receiver:



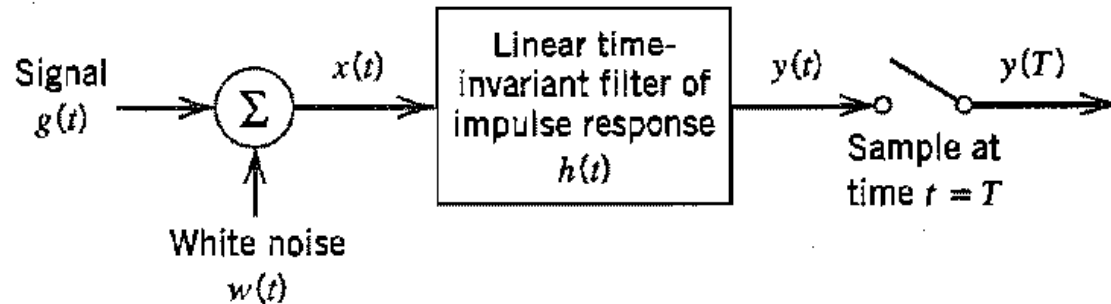
- At the filter input we have an impulse $g(t)$ corrupted with additive noise $w(t)$:
 - $x(t) = g(t) + w(t)$, $0 \leq t \leq T$
 - T – arbitrary observation interval
 - $g(t)$ – represents a binary 0 or 1 symbol
 - $w(t)$ – represents the random process of white noise with mean 0 and power spectral density equal to $N_0/2$

Matched Filter



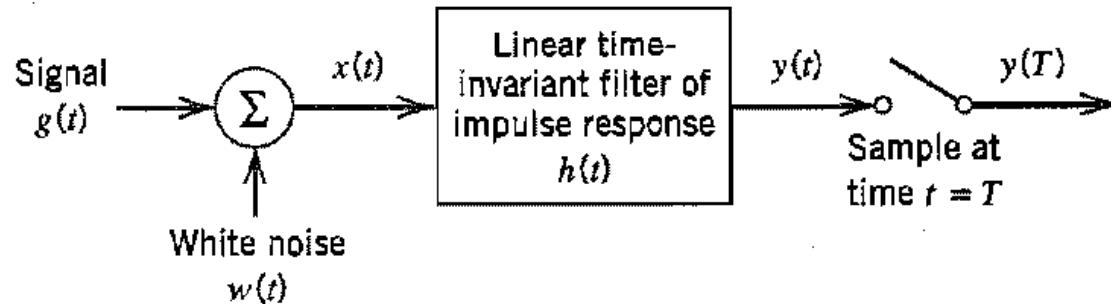
- The receiver is assumed to know the shape of the pulse $g(t)$
 - Thus, the only source of uncertainty is noise $w(t)$
- The function of the receiver is to optimally detect the pulses $g(t)$ at the expense of the received pulses $x(t)$
- The filter must be designed to minimize the effect of noise on its output

Matched Filter



- As the filter is linear, $y(t)$ is given by:
 - $y(t) = g_0(t) + n(t)$
 - $g_0(t)$ and $n(t)$ correspond to the input $x(t)$ components of the signal and noise, respectively
- **Objective:**
 - Maximize the relationship between signal power and noise power at sampling time T

Matched Filter



- Equivalent to maximizing the *peak pulse signal-to-noise ratio*:

$$\eta = \frac{|g_o(T)|^2}{E[n^2(t)]}$$

- The numerator represents the instantaneous power of the input signal and the denominator the expected value of the noise power

Matched Filter

- Conclusion:
 - How to specify the impulse response of the $h(t)$ filter in order to maximize the previous expression of the *peak pulse signal-to-noise ratio*?
- Being:
 - $G(f)$ – Fourier transform of $g(t)$
 - $H(f)$ – Filter frequency response
 - $G(f).H(f)$ – Fourier transform of $g_o(t)$
- The signal at the filter output (no noise) is given by the inverse Fourier transform of $G(f).H(f)$:

$$g_o(t) = \int_{-\infty}^{\infty} H(f)G(f) \exp(j2\pi ft) df$$



Matched Filter

- When at the output of the filter we sample at instant T :

$$|g_o(T)|^2 = \left| \int_{-\infty}^{\infty} H(f)G(f) \exp(j2\pi fT) df \right|^2$$

- Let us now consider the effect of noise $w(t)$ on the output:

$$S_N(f) = \frac{N_0}{2} |H(f)|^2$$

- The average value of the noise power $n(t)$ is:

$$\begin{aligned} E[n^2(t)] &= \int_{-\infty}^{\infty} S_N(f) df \\ &= \frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 df \end{aligned}$$



Matched Filter

- Substituting in :

$$\eta = \frac{|g_o(T)|^2}{E[n^2(t)]}$$

- the result is:

$$\eta = \frac{\left| \int_{-\infty}^{\infty} H(f) G(f) \exp(j2\pi f T) df \right|^2}{\frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 df}$$

- **Problem**: For a given $G(f)$ what is the frequency response of the filter $H(f)$ that maximizes η ?



Matched Filter

- To solve this optimization problem, we use the *Schwartz* inequality applied to the numerator of the previous expression:
 - Given 2 complex functions with real variable x , which satisfy the conditions:

$$\int_{-\infty}^{\infty} |\phi_1(x)|^2 dx < \infty \quad \int_{-\infty}^{\infty} |\phi_2(x)|^2 dx < \infty$$

- So:

$$\left| \int_{-\infty}^{\infty} \phi_1(x) \phi_2(x) dx \right|^2 \leq \int_{-\infty}^{\infty} |\phi_1(x)|^2 dx \int_{-\infty}^{\infty} |\phi_2(x)|^2 dx$$



Matched Filter

- The previous expression is valid for:

$$\phi_1(x) = k\phi_2^*(x)$$

- Where k is an arbitrary constant and the asterisk stands for the conjugate complex

- Making the change of variable:

$$\phi_1(x) = H(f) \text{ and } \phi_2(x) = G(f) \exp(j\pi fT)$$

- The numerator of η is:

$$\left| \int_{-\infty}^{\infty} H(f)G(f)e^{j2\pi fT} df \right|^2 \leq \int_{-\infty}^{\infty} |H(f)|^2 df \int_{-\infty}^{\infty} |G(f)|^2 df$$

Matched Filter

- Finally η is given by:

$$\eta \leq \frac{2}{N_0} \int_{-\infty}^{\infty} |G(f)|^2 df$$

- η does not depend on the frequency response of the filter $H(f)$
 - depends on signal energy and noise spectral density
- Thus, η will be maximum when $H(f)$ is chosen so that the equality is valid:

$$\eta_{\max} = \frac{2}{N_0} \int_{-\infty}^{\infty} |G(f)|^2 df$$



Matched Filter

- Using:

$$\phi_1(x) = k\phi_2^*(x)$$

- To find the optimal value of $H(f)$ we have:

$$H_{\text{opt}}(f) = kG^*(f) \exp(-j2\pi fT)$$

- $G^*(f)$ represents the conjugate complex Fourier transform of $g(t)$ and k an appropriate scaling factor
- Specifies the optimal filter in the frequency domain
- minus the factor $\exp(-j2\pi fT)$, the frequency response of the optimal filter is the same as the complex conjugate Fourier transform of the input signal!



Matched Filter

- To characterize the impulse response of the optimal filter in the time domain, the inverse Fourier transform is applied:

$$h_{\text{opt}}(t) = k \int_{-\infty}^{\infty} G^*(f) \exp[-j2\pi f(T - t)] df$$

- For a real signal $g(t)$ we have that $G^*(f) = G(-f)$, so:

$$\begin{aligned} h_{\text{opt}}(t) &= k \int_{-\infty}^{\infty} G(-f) \exp[-j2\pi f(T - t)] df \\ &= k \int_{-\infty}^{\infty} G(f) \exp[j2\pi f(T - t)] df \\ &= kg(T - t) \end{aligned}$$



Matched Filter Properties

- The matched filter's impulse response is a time-reversed and time-delayed version of the input $g(t)$:

$$h_{\text{opt}}(t) = kg(T - t)$$

- In other words, the filter impulse response is defined only by the input signal waveform $g(t)$ (unless the scale factor k and the delay T)
- The frequency response of the matched filter (less than the delay T) is the complex conjugate of the Fourier transform of the input signal $g(t)$

$$H_{\text{opt}}(f) = kG^*(f) \exp(-j2\pi fT)$$



Matched Filter Properties

- The *peak pulse signal-to-noise ratio* η depends only on the signal energy and the power spectral density of the white noise at the output of the filter (see Haykin demo):

$$\eta_{\max} = \frac{(kE)^2}{(k^2 N_0 E / 2)} = \frac{2E}{N_0}$$

- From the previous equation it can be seen that the maximum value of η does not depend on the input waveform
 - All signals with the same energy are equally effective

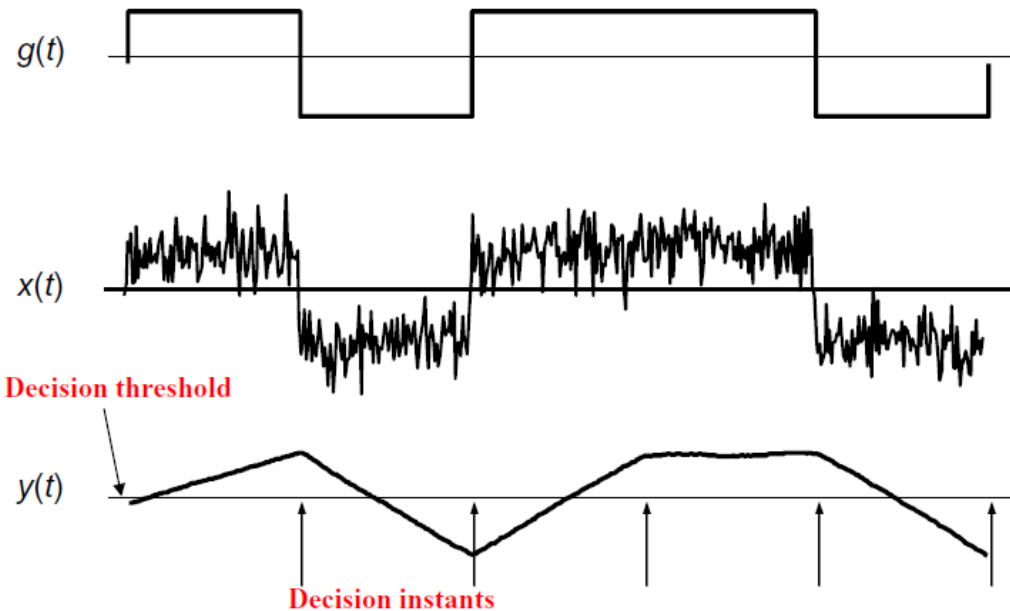
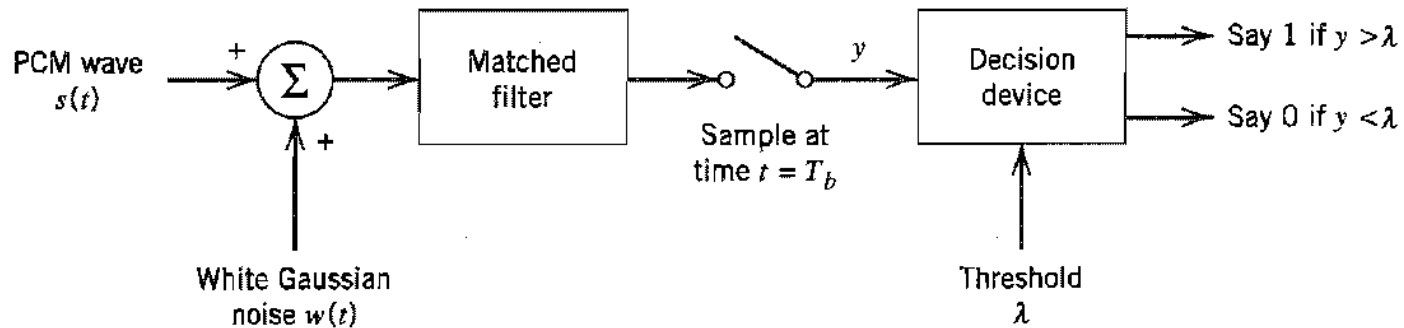


Matched Filter Properties

- A matched filter is a linear and invariant system whose main function is to detect the presence of a signal with a known waveform
- The output signal of the filter will be the correlation of the reference signal with the received signal with noise
- This procedure is equivalent to convoluting the received signal with a delayed version of the signal, which is used as a reference
- The matched filter is the optimal system to maximize SNR in the presence of Additive White Gaussian Noise (AWGN)



Matched Filter



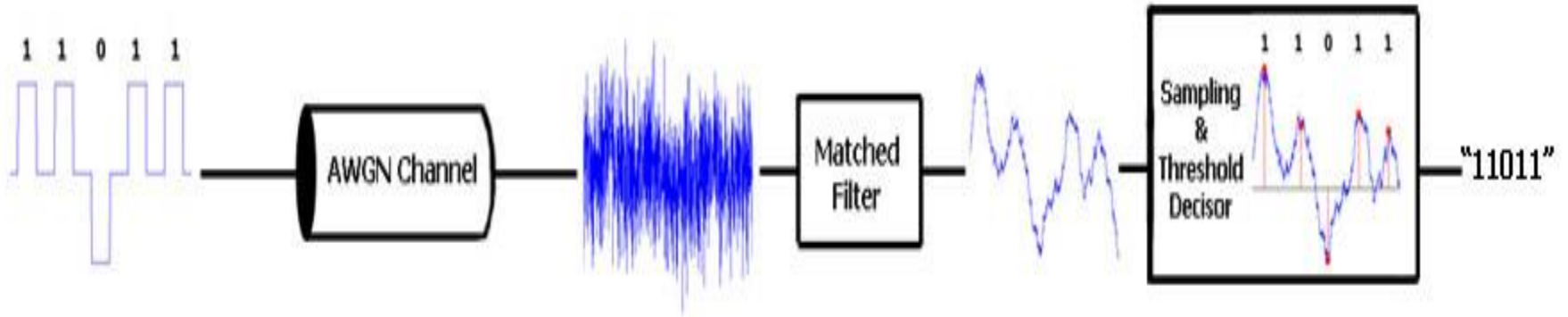
Matched Filter

- This operation is similar to that of not using a matched filter, with the essential difference that in this case the decision was made on the basis of a single sample of the signal affected by noise
- When using the matched filter, the decision is based on the result of the signal integration, that is, on the signal area in the transmission period of a symbol
 - It is easy to see that a single sample is much more likely to give rise to a wrong decision than the area of the signal
- As the signal area allows checking whether the signal is predominantly above or below the decision level, it is not just a sample that determines the decision, but the entire transmission range of a symbol



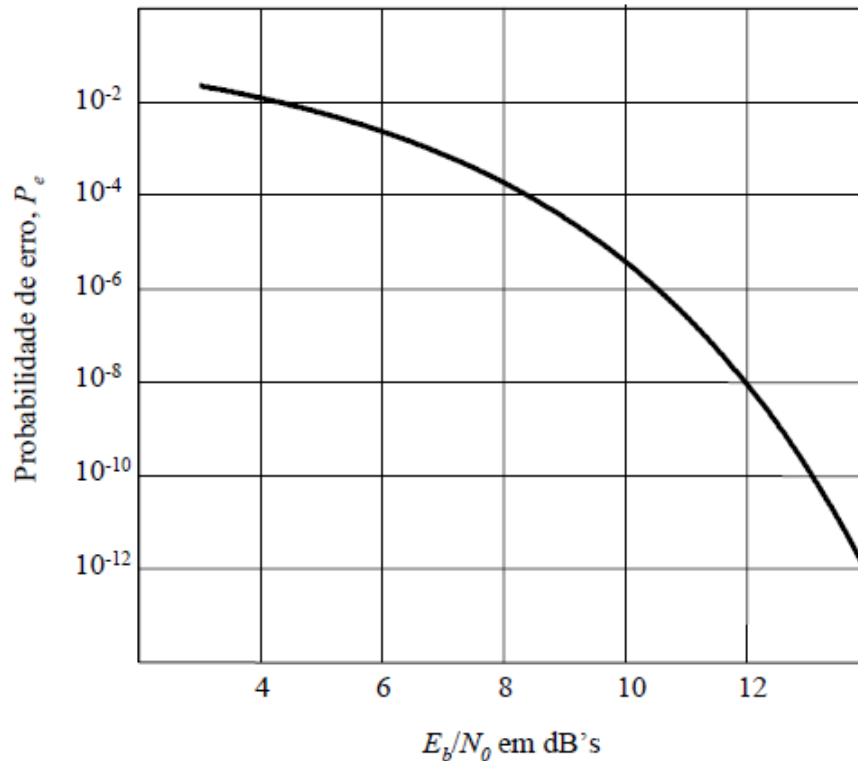
Matched Filter

- Overview:



Matched Filter

- Error probability as a function of the ratio between bit energy and noise power spectral density using a matched filter in an NRZ digital transmission with bipolar pulses



$$P_e = \frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{E_b}{N_0}} \right)$$

E_b – bit energy, that is, $E_b = A^2 T$

Non-Regenerative and Regenerative Repeaters

- In point-to-point transmission, to compensate for losses, repeaters are used



- In digital systems, repeaters can be of two types:

- Non-Regenerative Repeaters

- only amplify the signal
- noise is cumulative



$$P_e = \frac{1}{2} \operatorname{erfc} \left[\sqrt{\frac{1}{m} \left(\frac{E_b}{N_0} \right)} \right]$$

- Regenerative Repeaters

- amplify and regenerate the digital signal
- the probability of error is cumulative
- lead to better overall performance



$$P_e \approx m \frac{1}{2} \operatorname{erfc} \left[\sqrt{\left(\frac{E_b}{N_0} \right)} \right]$$

Non-Regenerative Repeaters

- The hops are assumed to be identical
 - situation that occurs in practice, in general
- In the case of non-regenerative repeaters, the noise is accumulated along the link, that is, the total noise will be:

$$N = \sum_{i=1}^m N_i = mN_1$$

- where N_1 is the noise on a single link and m the number of hops
- There will be regeneration only at the end of the chain receiver, with the total E_b/N_0 ratio and the resulting error probability expressed as a function of the $(E_b/N_0)_1$ ratio for a single hop:

$$\frac{E_b}{N_0} = \frac{1}{m} \left(\frac{E_b}{N_0} \right)_1 \Rightarrow P_e = \frac{1}{2} \operatorname{erfc} \left[\sqrt{\frac{1}{m} \left(\frac{E_b}{N_0} \right)_1} \right].$$



Regenerative Repeaters

- In the case of regenerative repeaters, at the end of each hop the digital signal is recovered, possibly with errors, and is sent again
 - errors accumulate at the end of the link
- Total error probability:

$$P_e \approx \sum_{i=1}^m P_{ei} = mP_{e1} = m \frac{1}{2} \operatorname{erfc} \left[\sqrt{\left(\frac{E_b}{N_0} \right)_1} \right]$$

- Where P_{e1} is the error probability associated with a single hop
- Note that this last expression is approximate since the occurrence of multiple errors is neglected, which is valid, considering that the error probability of each hop is always very low

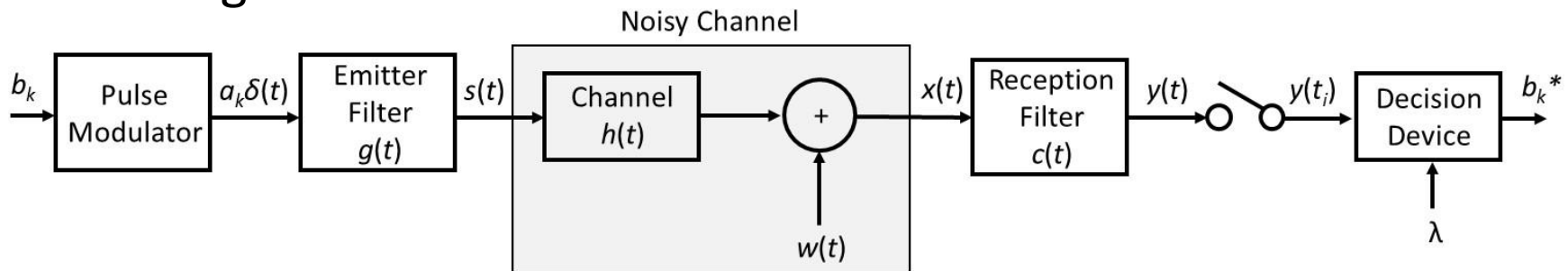
Regenerative Repeaters

- Conclusion:
 - As the error probability increases very quickly with decreasing E_b/N_0 , systems with regenerative repeaters perform better than systems with non-regenerative repeaters, requiring less signal power output for the same error probability



Intersymbolic Interference

- Most serious cause of errors in baseband digital transmission
 - It can be controlled by introducing suitable filters in the emission and reception
- Due to the fact that the impulse response of the real channels is “long”, the response to a pulse influences the reception of the following



b_k - Binary data sent: “1” and “0”

a_k - Polar data sent: “+1” and “-1”

$s(t)$ - Pulses formatted by the emission filter for the transmission (T_b)

$x(t)$ - Received Pulses with AGWN - $w(t)$

$y(t)$ - Pulses formatted by the reception filter to reduce intersymbolic interference

$y(t_i)$ - Impulses sampled at time instants $t_i = i \cdot T_b$

b_k^* - Binary data received



Intersymbolic Interference

- The transmitted pulses will be given by :

$$s(t) = \sum_{k=-\infty}^{\infty} a_k g(t - kT_b)$$

- where a_k corresponds to symbols (1 for bit “1” and -1 for bit “0”) and T_b is the period of a symbol

- The signal at the input of the sampler is given by:

$$y(t) = \mu \sum_{k=-\infty}^{\infty} a_k p(t - kT_b) + n(t)$$

- where $p(t)$ is the waveform of each pulse, μ is a gain resulting from normalization and $n(t)$ is the noise filtered by the reception filter

Intersymbolic Interference

- $p(t)$ results jointly from the characteristics of the emission filter, the channel and the reception filter, since:

$$\mu P(f) = G(f)H(f)C(f)$$

- At the output of the sampler, the signal sampled at time $t_i = i.T_b$ has the following form :

$$y(t_i) = \mu \sum_{k=-\infty}^{\infty} a_k p[(i-k)T_b] + n(t_i) = \mu a_i + \mu \sum_{\substack{k=-\infty \\ k \neq i}}^{\infty} a_k p[(i-k)T_b] + n(t_i)$$

- Therefore, if there is no intersymbolic interference and there is no noise, we will have at the sampler output:

$$y(t_i) = \mu a_i$$



Intersymbolic Interference

- To ensure that there is no intersymbolic interference, it is necessary that:

$$p(iT_b - kT_b) = \begin{cases} 1, & i = k \\ 0, & i \neq k \end{cases}$$

- Requires that, for a channel with a given impulse response $h(t)$, it is necessary to choose the appropriate filters for the emitter and receiver
- The previous condition translates into the frequency domain by the following condition:

$$\sum_{n=-\infty}^{\infty} P(f - nR_b) = T_b \quad R_b = \frac{1}{T_b}$$

- Where $R_b = 1/T_b$ is the binary throughput. This condition for there to be no intersymbolic interference is called the “Nyquist criterion”

Intersymbolic Interference

- One way to satisfy the previous condition is to make $P(f)$ equal to:

$$P(f) = \begin{cases} 1/2W, & -W < f < W \\ 0, & |f| > W \end{cases} \quad W = \frac{R_b}{2} = \frac{1}{2T_b}$$

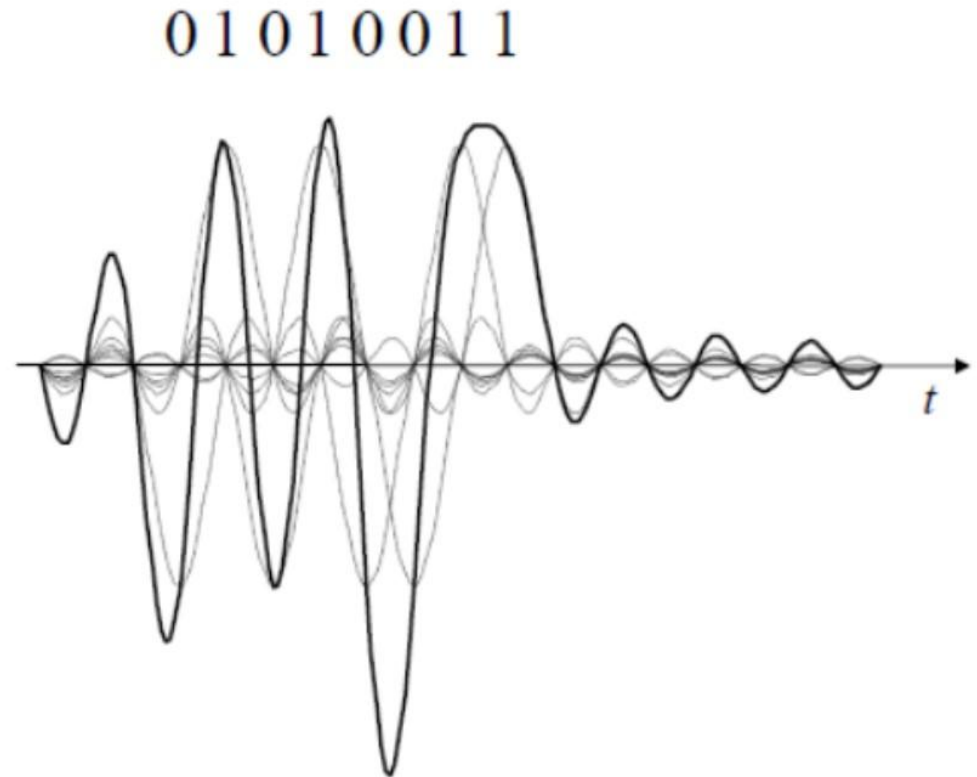
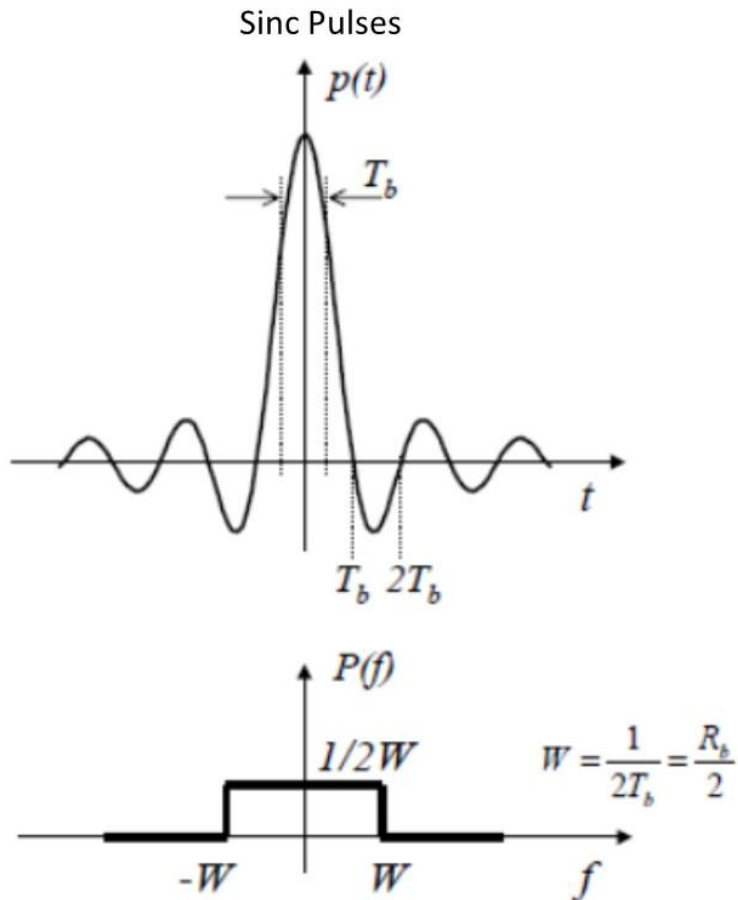
- which corresponds to pulses with the shape of a cardinal sine (sinc function):

$$p(t) = \frac{\sin(2\pi Wt)}{2\pi Wt}$$

- The bit rate $R_b = 2W$ is called “Nyquist throughput” and W is called “Nyquist bandwidth”
- For a given bandwidth W the maximum possible bit rate without intersymbolic interference is R_b , provided that cardinal sine pulses are used



Intersymbolic Interference



Example of a transmission sequence without intersymbolic interference at sampling instants

Pulse Shaping

- Process of changing the waveform of transmitted pulses
- The objective is to make the transmitted signal more suitable for the communication channel, limiting the effective bandwidth of the transmission
- By filtering the transmitted pulses, the intersymbolic interference caused by the channel can be controlled
- Typically occurs after line encoding and modulation



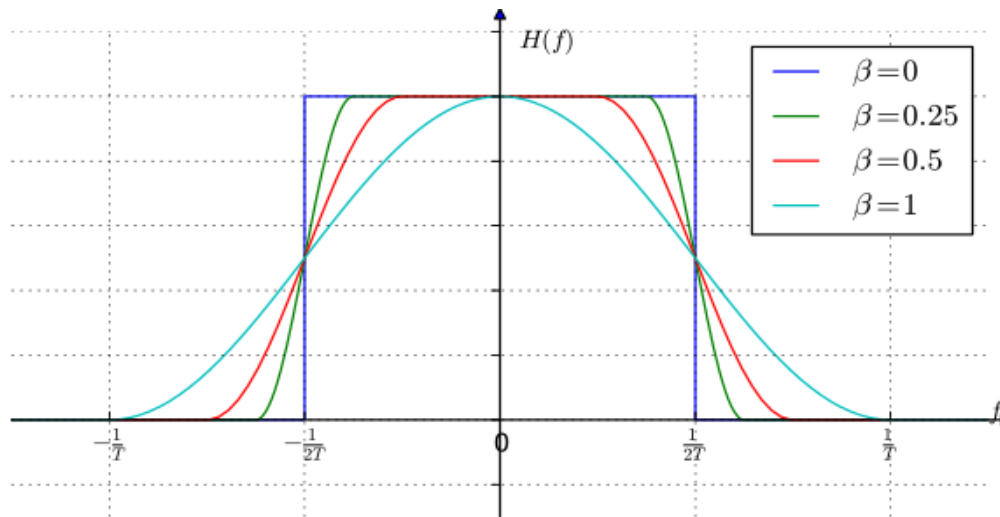
Pulse Shaping filters

- Cardinal Sine Filter
 - Also called a *Boxcar filter*, as its frequency response is rectangular in shape
 - Theoretically it is the best filter but it cannot be implemented precisely - non-causal filter
 - It is also problematic from a synchronization point of view, as a phase error results in intersymbolic interference
- Gaussian Filter
 - Produces an output pulse in the form of a Gaussian function



Pulse Shaping filters

- Raised-cosine filter
 - Practical to implement and widely used
 - Allow configurable bandwidth
 - We can choose a compromise between simplicity of implementation and spectral efficiency



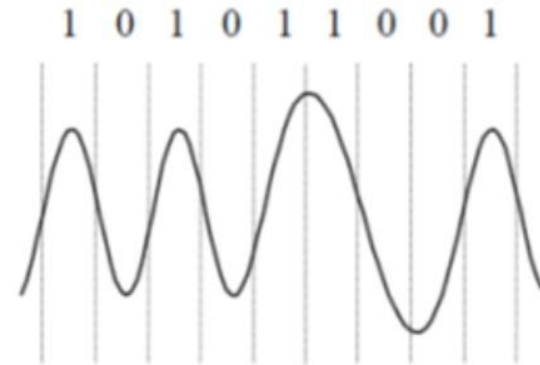
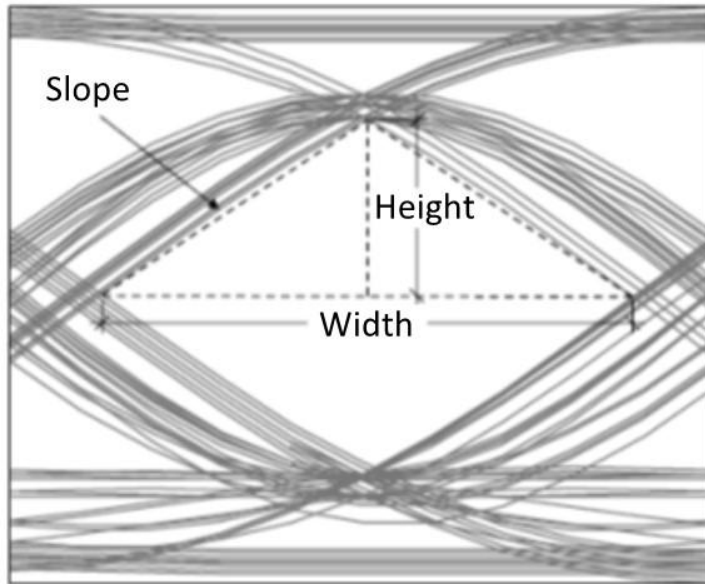
Raised cosine response to various roll-off factors

Eye Diagram

- Allows a qualitative experimental evaluation of the error probability
- One way to assess the effects of intersymbolic interference and noise is to superimpose all possible forms of the impulse on reception for the period of a symbol and interpret the diagram obtained
- Diagrams of this type are called “eye diagrams” because of their shape
- The diagram is built on the basis of the clock recovered at the reception



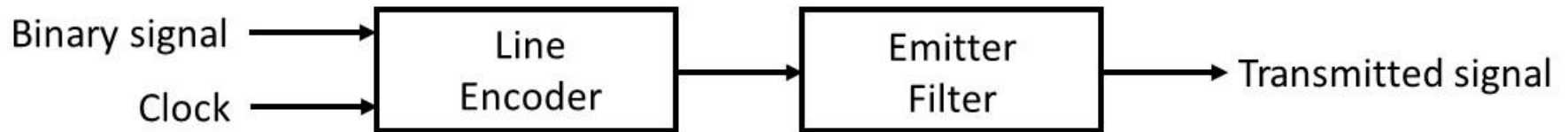
Eye Diagram



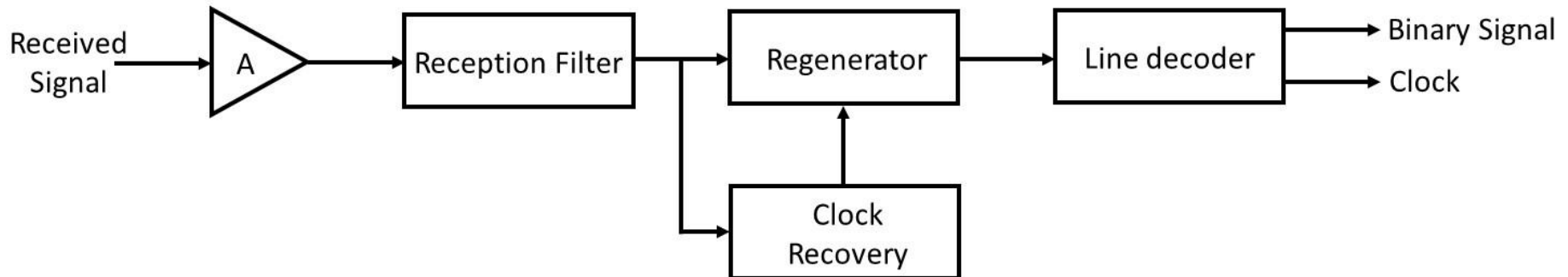
- From the eye diagram, several useful information can be extracted:
 - the width of the “eye” defines the time interval where sampling can be carried out without there being a decision error
 - the slope makes it possible to assess the system's sensitivity to errors in choosing the sampling time

Emitters and Receivers

- Emitter:



- Receiver:



Emitters and Receivers

- The transmitted signal comprises:
 - Line encoder: binary conversion - transmission code
 - not needed on repeaters where, between reception and emission, the signal is not converted to the original binary format
 - Emitter filter: generation of pulses with a format suitable for transmission, with a smaller band than rectangular pulses, while ensuring low intersymbolic interference



Emitters and Receivers

- Signal reception includes the following functions:
 - Amplification (A)
 - Receiver filter: signal conditioning for transmission compensation
 - performing, in particular, channel amplitude and phase distortion equalization, adaptive filtering and intersymbolic interference reduction;
 - Clock recover: obtaining the emitter clock by filtering the signal;
 - Regenerator: quantification of the signal in time (sampler) and in amplitude (decision device);
 - Line decoder: transmission code - binary conversion (not needed on repeaters)



Reading Material

- *Simon Haykin, Communication Systems, McMaster Univ, 5th Edition, 2009 (ISBN 978-047-16979-0-9)*
- *A. Bruce Carlson and Paul Crilly. Communication Systems – An Introduction to Signals and Noise, McGraw-Hill, 5th edition, 2009 (ISBN 978-007-338-040-7)*
- *Bernard Sklar and Fredric Harris, Digital Communications: Fundamentals and Applications, Pearson, 3rd Edition, 2020 (ISBN-13: 978-0134588568)*



Reading Material

